

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

**STUDENT
NAME**

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CLASS

2T

**INDEX
NUMBER**

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BIOLOGY

Paper 2 Core Paper

9648/02

FRIDAY 23 AUG 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

SECTION A]

Answer **all** questions.

SECTION B]

Answer any **one** question.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
Section A	80
1 [14]	
2 [10]	
3 [12]	
4 [12]	
5 [10]	
6 [12]	
7 [10]	
Section B	20
8 or 9 (a)	
(b)	
(c)	

Section A

Answer **all** the questions in this section.

- 1 The peptidoglycan cell wall of a prokaryote, *Escherichia coli*, is made up of two different sugars: N-acetylmuramate (NAM) and N-acetylglucosamine (NAG). NAM and NAG have different groups being attached to the C3 position.

Fig.1.1 shows the molecular structure of NAM and NAG.

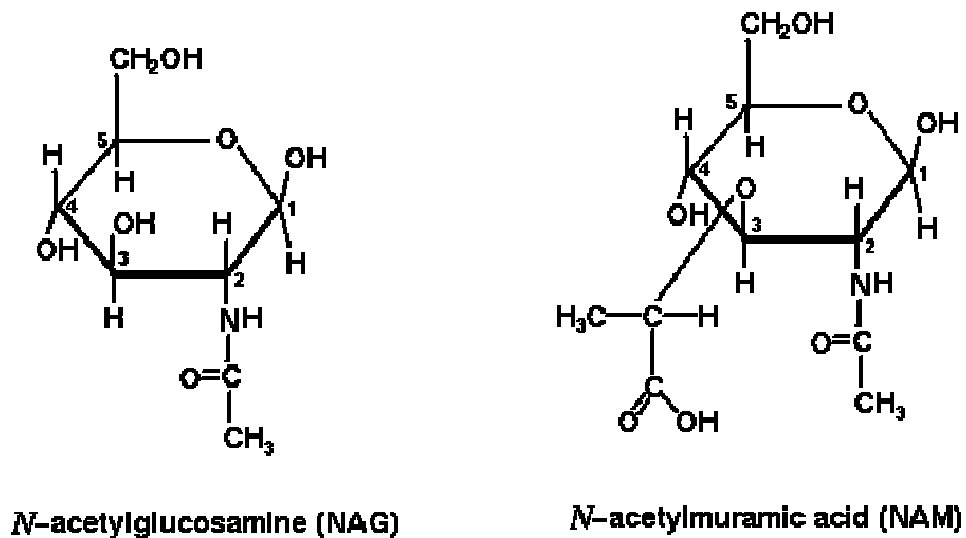


Fig. 1.1

Fig.1.2 shows the structure of the peptidoglycan cell wall of *Escherichia coli*.

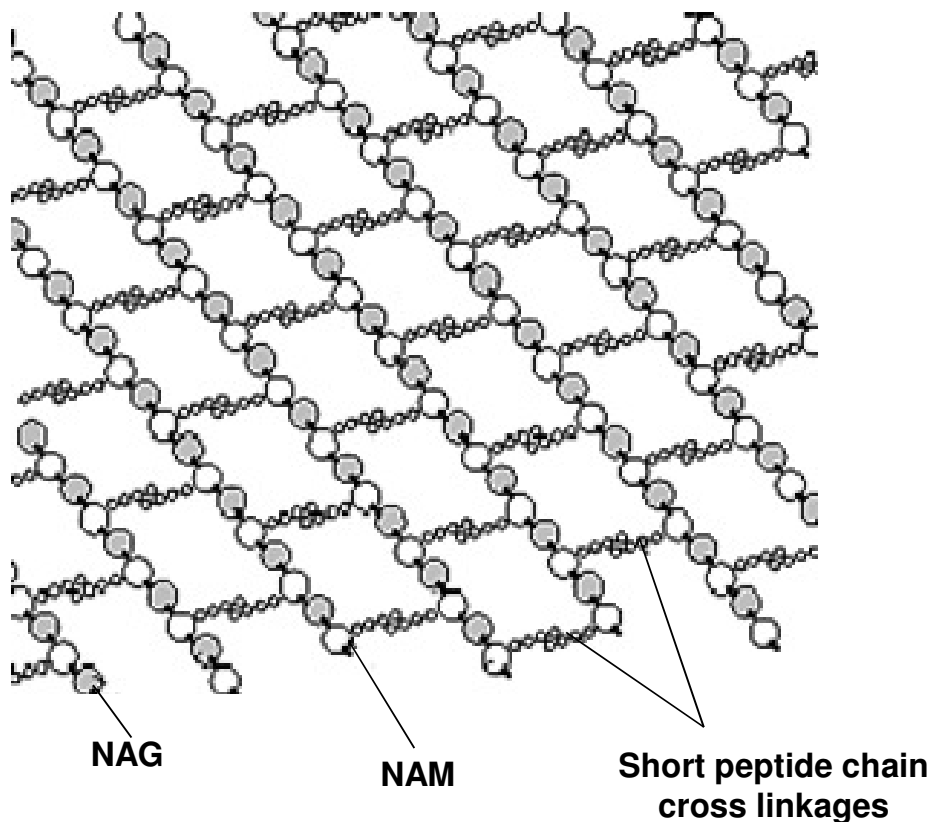


Fig. 1.2

- (a) With reference to Fig. 1.1, suggest the name of the bond that will be formed between the 2 sugars, NAG and NAM.

..... [1]

- (b) With reference to Fig.1.1 and Fig. 1.2,

- (i) distinguish between bacterial peptidoglycan and cellulose found in the plant cell wall.

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..... [3]

- (ii) suggest why peptidoglycan associates with amino acids.

.....
..... [1]

Lysozyme is considered antibacterial as it degrades polysaccharides that are found in the cell wall of many bacteria, including that of *Escherichia coli*. It is a globular protein with a deep cleft across a part of its surface.

Fig. 1.3 shows the action of part of lysozyme to the cell wall of *Escherichia coli*. Numbers 1 to 6 represent the number of NAG or NAM that are bound to the active site of lysozyme.

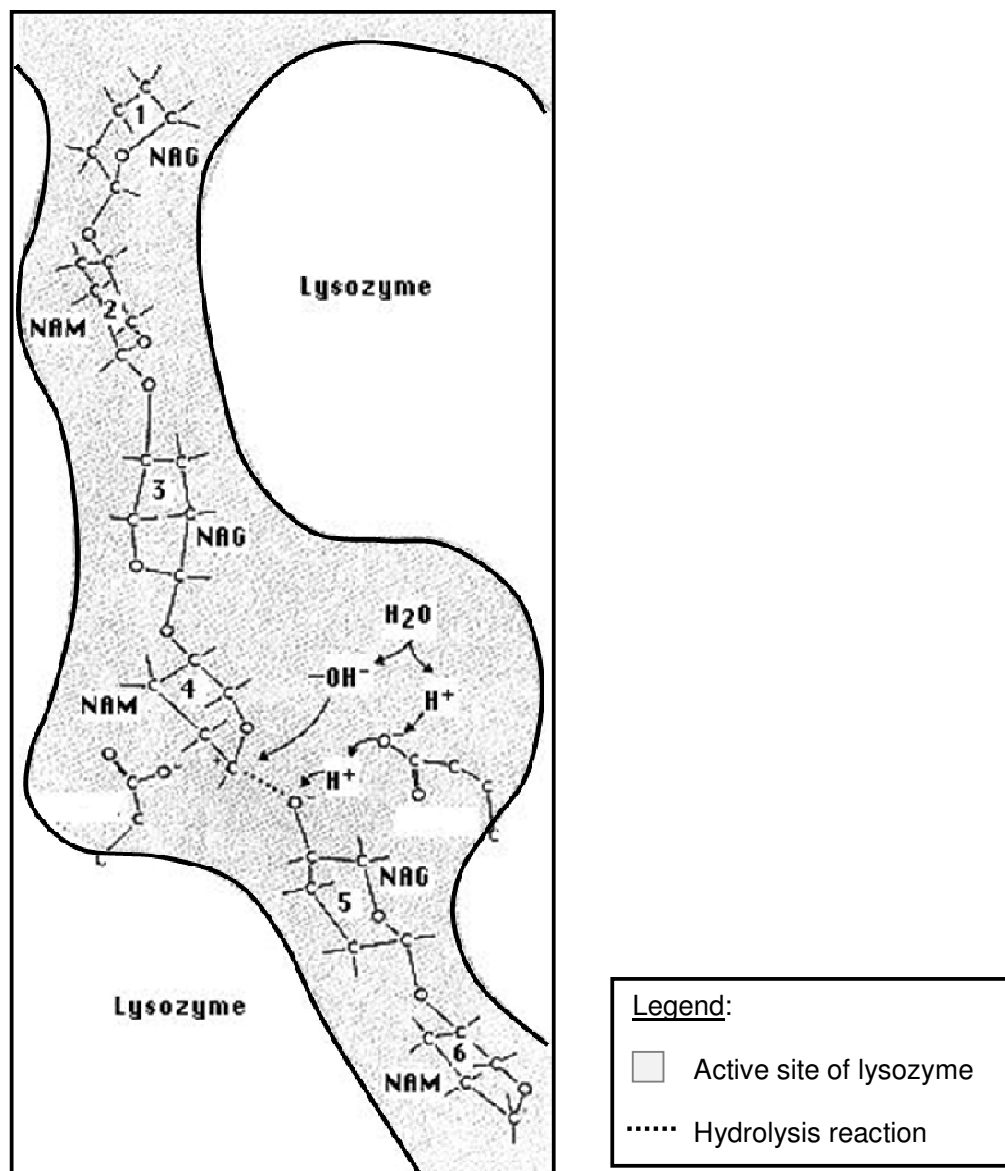


Fig. 1.3

(c) With reference to Fig. 1.3, suggest how lysozyme performs its anti-bacterial function.

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..... [3]

Fig.1.4 shows the effect of temperature on lysozyme activity.

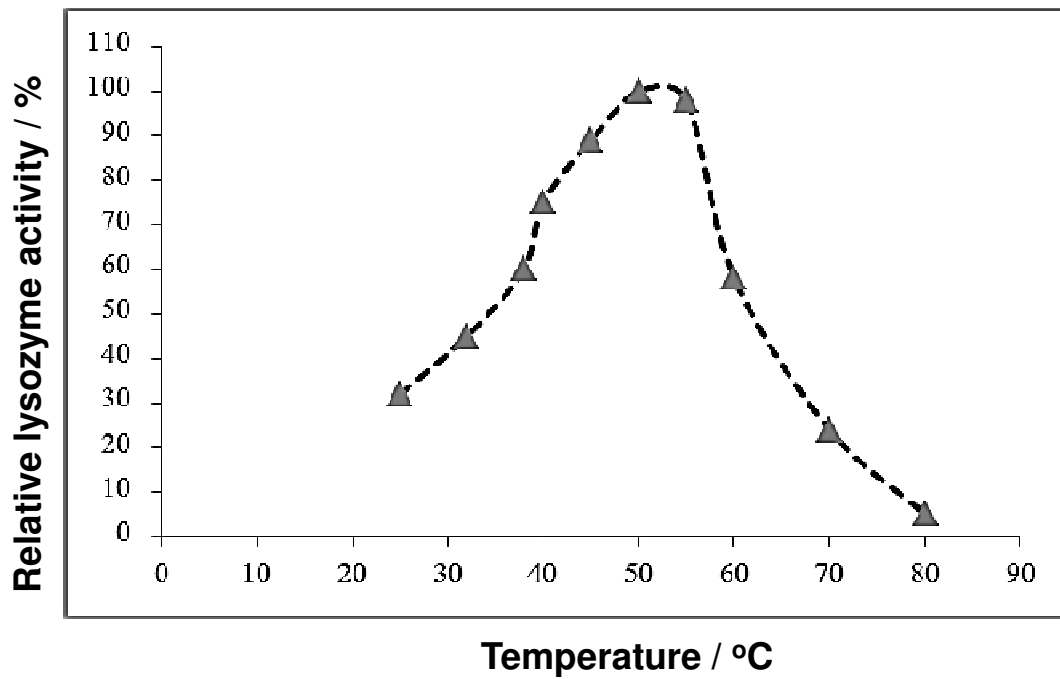


Fig. 1.4

(d) With reference to Fig.1.4,

(i) describe the effect of temperature on relative lysozyme activity.

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..... [3]

(ii) explain why there is a decrease in relative lysozyme activity at temperature above 50°C.

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..... [3]

[Total: 14]

- 2 (a) A clinic sequenced the β -globin gene locus of five different patients and tabulated the results in the following table.

Table 1. Genetic profile of the 6th codon of patients' β -globin gene

Patient	DNA codon sequence*	Change in amino acid
1	GAG	Glu (unchanged)
2	GAA	Glu (unchanged)
3	GTG	Glu $\rightarrow$ Val
4	GAC	Glu $\rightarrow$ Asp
5	GTG	Glu $\rightarrow$ Val

* DNA sequence on the 6th codon of the human β -globin gene

Based on the information in Table 1,

- (i) which patient(s) has/have the sickle-cell anaemia mutation?

..... [1]

- (ii) explain how the change of amino acid would result in a sickle shaped red blood cell.

.....

 [3]

- (iii) Suggest and explain why patient 4 does not have sickle cell anaemia.

.....

 [2]

Figure 2.1 shows a linear chromosome undergoing the first round of DNA replication.

The arrows show the direction of synthesis for the daughter DNA strands.

Ends A, B C and D represent the ends of the newly synthesized daughter DNA strands.

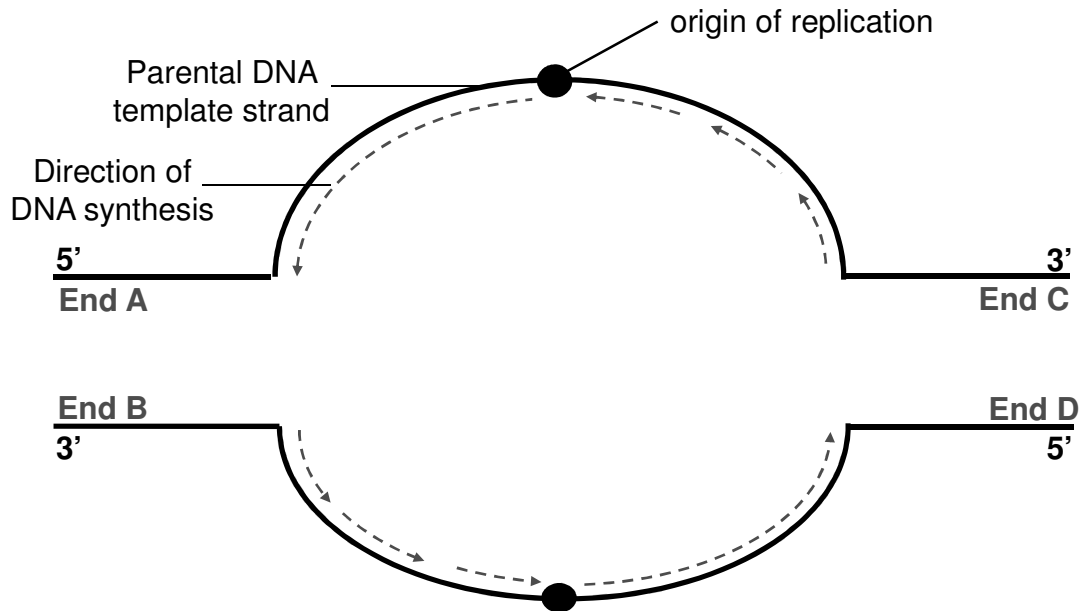


Fig. 2.1

(b) With reference to Fig. 2.1,

- (i) state which end(s) will experience the end-replication problem after the first round of DNA replication has been completed.

..... [1]

- (ii) suggest one modification that a researcher could do to the parental DNA strands in order to completely remove the end-replication problem.

.....
 [1]

(c) Telomerase is an enzyme that is found in certain cell types and it helps to ameliorate the end replication problem by lengthening the ends of the chromosomes, which are also known as telomeres.

Explain why the telomerase enzyme does not prevent the end-replication problem from occurring despite being able to lengthen the telomeres.

.....

 [2]

[Total: 10]

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3 The structural genes in the bacterial operon can be regulated by either the inducible or repressible system.

(a) Compare the two systems of regulation.

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..... [3]

(b) Suggest two advantages of regulating gene expression by the operon model.

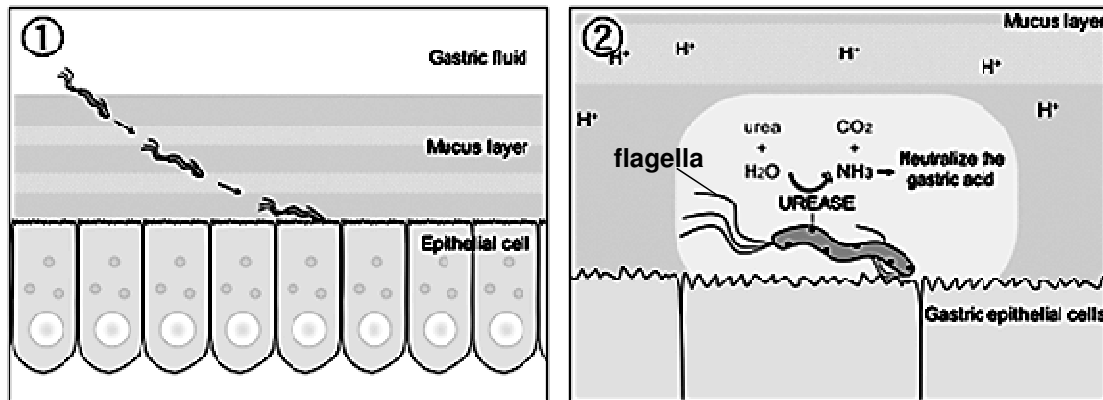
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..... [2]

Helicobacter pylori is a spiral shaped bacterium that infects the gastrointestinal tract. Epithelial cells lining the stomach secrete a thick layer of mucus to protect against digestive enzymes and hydrochloric acid. *H. pylori* is able to colonize the harsh environment of the stomach by swimming through this layer of mucus. The bacterium lives beneath the mucus layer, adjacent to the surface and infect gastric epithelial cells. Fig. 3.1 illustrates how *H. pylori* invades its host.



Source: <http://www.stanford.edu/group/parasites/ParaSites2006/Helicobacter/Introduction.html>

Fig. 3.1

- (c) With reference to Fig. 3.1, explain why *H. pylori* is able to colonize the gastric epithelial cells despite the “harsh environment of the stomach”.

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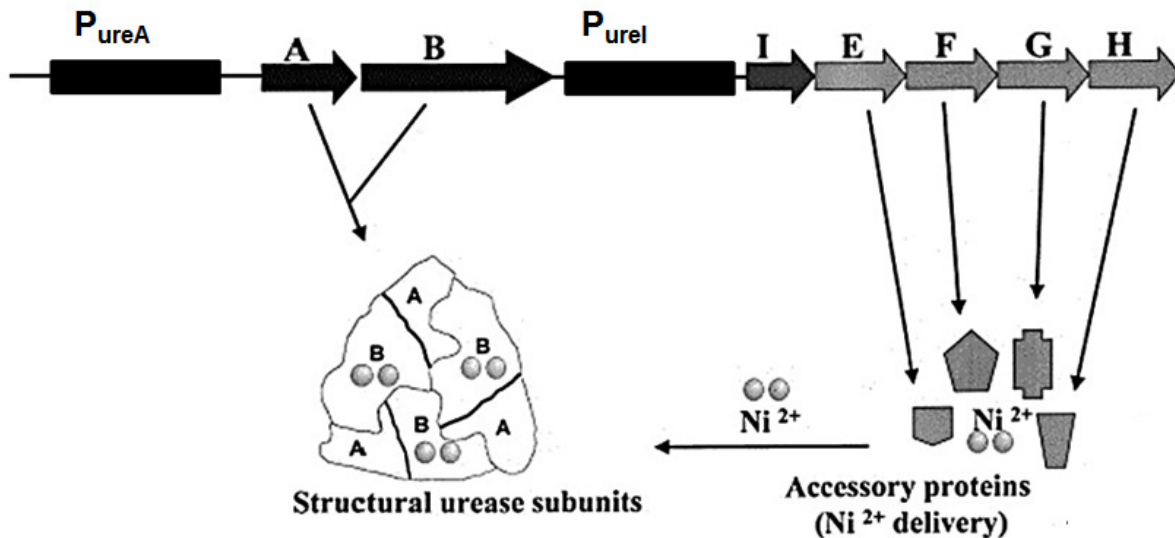
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The genes encoding the enzyme urease in *H. pylori* are located as a single gene cluster (as two operons) on the bacterial chromosome as shown in Fig. 3.2.

- Urease is a multimeric nickel-containing enzyme which consists of six UreA (A) and UreB (B) subunits.
- Two genes (*ureA*, *ureB*) are under the control of the P_{ureA} promoter, encode the catalytically inactive apoenzyme.
- Five genes (*ureI*, *ureE*, *ureF*, *ureG* and *ureH*) are under the control of the P_{ureI} promoter.
- The gene *ureI* encodes an acid-activated urea transporter.
- Accessory genes *ureE*, *ureF*, *ureG*, and *ureH* encode accessory proteins that aid in the insertion of nickel ions (Ni^{2+}) into the apoenzyme to form catalytically active urease.



Modified from Source: *Helicobacter pylori: Physiology and Genetics*. Mobley et. al. 2001

Fig. 3.2

(d) Suggest why it may be advantageous to have two operons controlling the expression of urease.

.....

 [1]

A team of researchers would like to study the effect of nickel (Ni^{2+}) availability on urease expression and urease activity.

Colonies of *H.pylori* were grown in unsupplemented media and media supplemented with 1 μM and 100 μM NiCl_2 (Nickel chloride). The level of mRNA (*ureA* and *ureB*) and urease activity of *H.pylori* bacteria grown in each media were measured and recorded in Table 3.3 and Figure 3.4, respectively.

Type of media	Change in relative level of mRNA (<i>ureA</i> and <i>ureB</i>)
Unsupplemented (Control)	+ 0.0
Media with 1 μM NiCl_2	+ 1.3
Media with 100 μM NiCl_2	+ 3.3

Table 3.3

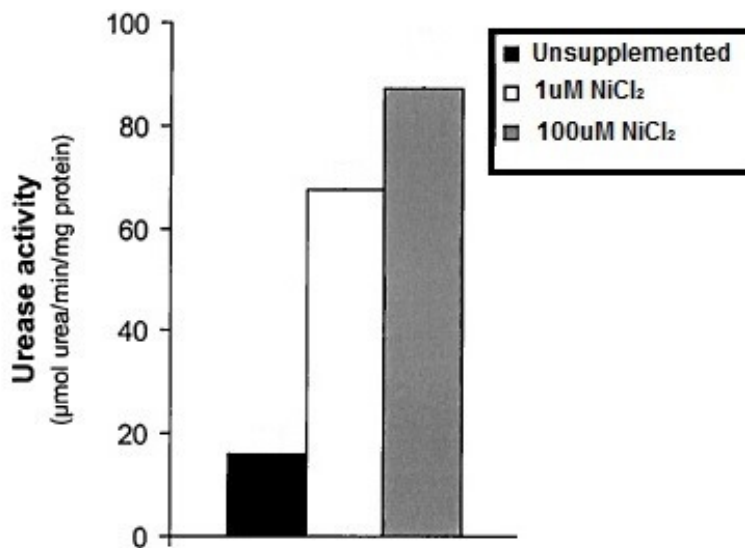


Fig. 3.4

(e) With reference to Table 3.3 and Fig. 3.4, suggest how nickel affects

(i) urease expression.

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..... [2]

(ii) urease activity.

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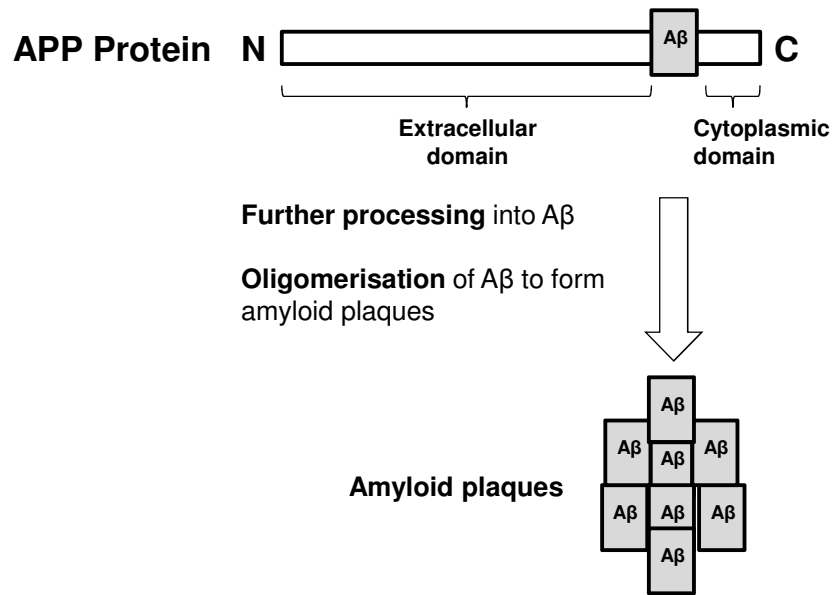
..... [2]

[Total: 12]

- 4 The *APP* gene is found in chromosome 21 and encodes for a protein known as amyloid precursor protein (APP). APP protein is involved in synapse formation and development of neurons. APP contains a $A\beta$ component that is likely to play a natural role in the ability of neurons to adapt and change over time (plasticity).

However, when the *APP* gene undergoes a mutation, an overproduction of $A\beta$ will occur and they will be released from such mutated cells. Excess $A\beta$ will accumulate in the brain and aggregate to form clumps known as amyloid plaques which are commonly found in patients afflicted with Alzheimer disease (AD). With progressive buildup of toxic plaques and the consequential death of neurons, patients will usually show worsening signs of AD.

Fig. 4.1 shows how amyloid plaques are formed.



Modified from Source: <http://www.sciencedirect.com/science/article/pii/S0361923011001870>

Fig. 4.1

- (a) With reference to Fig. 4.1, describe how APP is processed into $A\beta$.

.....

 [2]

Proteins with significantly different structures in various cells were found to have originated from the *APP* gene.

- (b) Describe and explain how such proteins could have arisen from a single gene.

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 [3]

(c) Describe how excess A β can be removed within a cell.

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..... [2]

To support studies in AD, researchers have isolated a DNA sequence which can be used to trigger the overexpression of *APP* gene. When this sequence was placed distally upstream of *APP* gene and then introduced into suitable host cells, the amount of A β secreted was significantly increased.

(d) Describe and explain the mechanism which altered the amount of A β synthesized.

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..... [3]

In another large scale study, autopsies were carried out to determine the presence of amyloid plaques in people across different age groups (as indicated by percentage positive). This data was analyzed together with the prevalence of AD-related dementia diagnosed in patients across different age groups (as indicated by percentage positive). The data is being presented in Fig. 4.2.

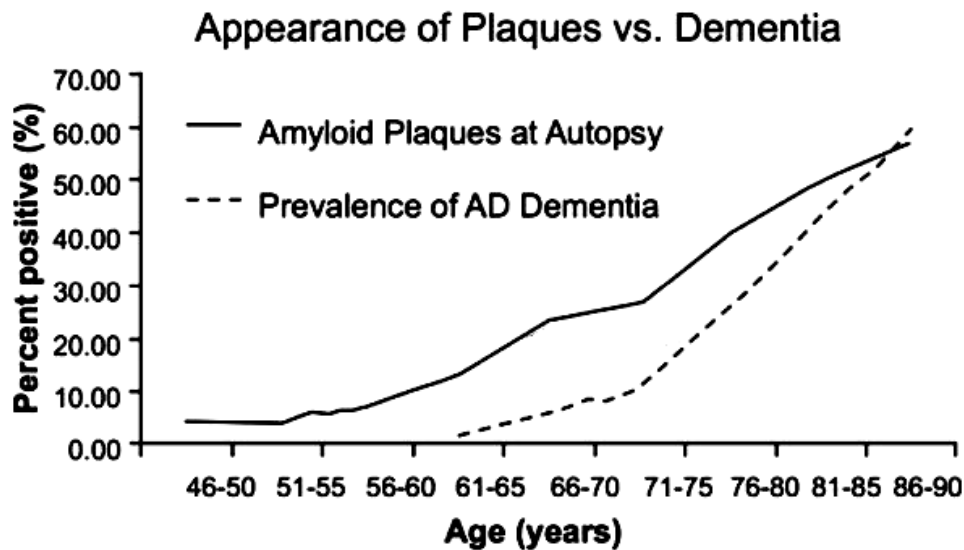


Fig. 4.2

(e) Discuss the significance of the results shown.

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..... [2]

[Total: 12]

- 5 (a) The Manx gene determines the presence or absence of tails in cats. The Manx gene is controlled by Gene S, which determines the development of the embryo cat's spine.

The allele S^N codes for normal spine, which results in a long tail being formed. S^A codes for an abnormal, short spine. Cats which are homozygous recessive for this gene will not have their spines developed and will die early in their development. Manx cats are tailless with a heterozygous genotype, as shown in Fig. 5.1.



Fig. 5.1

A second gene which controls for the appearance of the coat is located on the X chromosome. Allele D codes for 'dappled' coat (D) while allele d codes for 'plain' coat (d).

A male Manx cat with dappled coat was mated on several occasions to the same female and a large number of offspring consisted of the following phenotypes, as summarised in Table 5.1 was observed.

Table 5.1

Tail Type	Gender	Coat colour	% no. of kittens
Long tailed	Male	Dappled	12.5
Long tailed	Female	Dappled	12.5
Manx	Male	Dappled	25
Manx	Female	Dappled	25
Did not survive development			25

Using suitable symbols, draw a genetic diagram to explain the results in Table 5.1.

- (b) A farmer has two homozygous varieties of tomatoes. One variety, called *Little Pete*, has tomatoes with an average diameter of 2.5 cm. The other variety, *Big Boy*, has tomatoes with average diameter of 13.5 cm. The farmer performed a cross between *Little Pete* and *Big Boy*. The F_1 generation was then interbred.

Fig. 5.2 shows the diameters of the tomatoes in each generation.

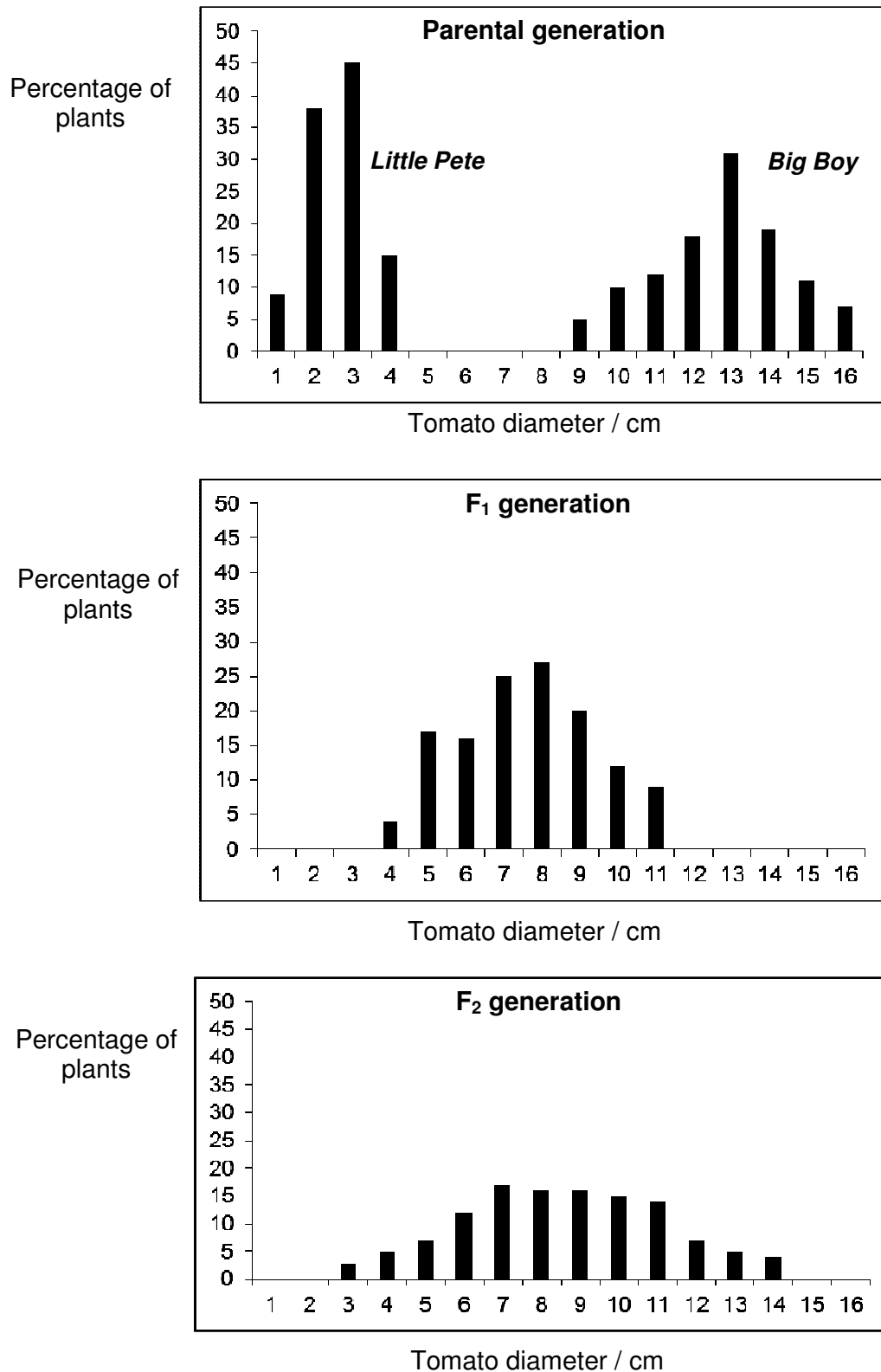


Fig. 5.2

(i) With reference to Fig. 5.2, comment on the variation seen in the F_1 generation.

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(ii) Explain why the variation observed of the F_2 generation is greater than that of the parental generation and the F_1 generation.

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..... [3]

[Total: 10]

- (ii) describe how glucagon increases the blood glucose level when it is bound to its receptor in effector cells.

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..... [5]

- (iii) explain how signal amplification occurs.

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..... [2]

- (c) A group of liver cells has been found to contain a mutation in a gene encoding for a specific phosphatase, **phosphatase W**. Phosphatase W has been known to act on a particular substrate in the glucagon signalling pathway to prevent glycogenolysis.

With reference to Fig. 6.1, describe how phosphatase W prevents glycogenolysis.

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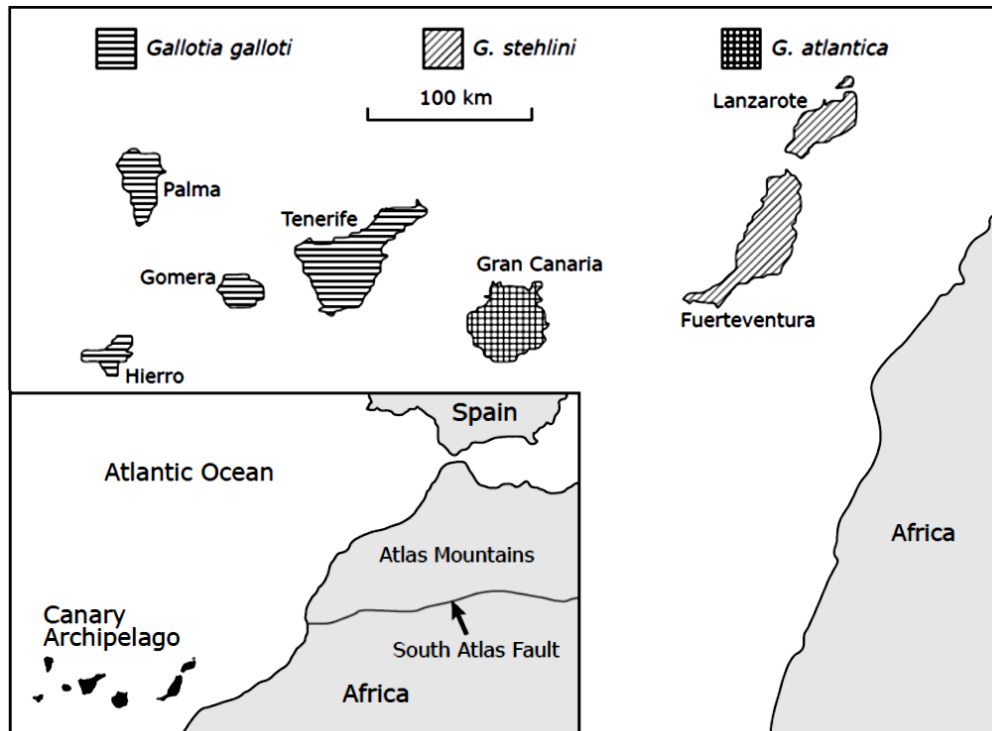
..... [3]

[Total: 12]

- 7 The Canary Islands form an archipelago of seven volcanic islands just west of the African continent. The island chain starts about 85 km west of the continent, following a fault line of the Atlas Mountains in northern Africa. Geologists theorize that a geologic hot spot of upwelling magma has been drifting westward for the past 20 million years, gradually forming the islands as it moves. Volcanic islands are particularly good laboratories for evolutionary science because they can be dated accurately using radioactive isotope decay and because they start out as lifeless masses of rock emerging from the sea. Today, several distinct species of lizards, namely, *Gallotia atlantica*, *G. stehlini*, and *G. galloti*, colonise these islands.

Fig. 7a Maximum age of the Canary Islands in millions of years. (*Anguita et al.*, 1986)

Lanzarote & Fuerteventura	Gran Canaria	Tenerife	Gomera	Palma	Hierro
24.0	17.1	15.1	5.3	2.0	0.8



(Redrawn from Thorpe et al., 1993) Adapted.

Fig. 7b. Distribution of *Gallotia atlantica*, *Gallotia stehlini*, and *Gallotia galloti*.

G. galloti has colonized the four westernmost islands and each population is morphologically distinct from the others.

- (a) Given that the lizards are distinct from their mainland counterparts, state and define the kind of speciation that was involved in the colonisation of the Canary Islands?

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 [2]

- (b) Aside from speciation, how do these lizards remain genetically distinct from their mainland counterparts?

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 [1]

(c) Suggest how the *Gallotia* lizards first colonised the Canary Islands.

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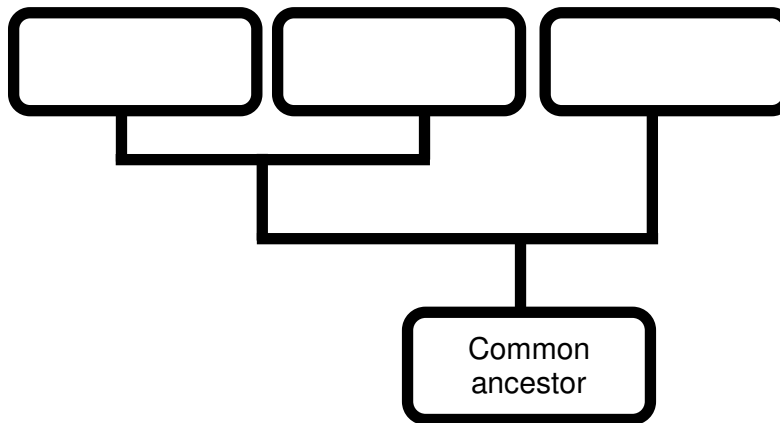
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..... [2]

(d) With reference to Fig. 7a and Fig. 7b,

(i) propose a sketch of the phylogenetic relationship between species. *Gallotia atlantica*, *Gallotia stehlini* and *Gallotia galloti*.



[1]

(ii) explain your answer in (d)(i).

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..... [2]

(e) The survival of the organism according to Darwin depends much on the selective advantage of certain gene traits that allow it an edge over others. Define neutral theory and explain how neutral theory counters Darwin's theory of natural selection.

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..... [2]

[Total: 10]

Section B

Answer **one** question.

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in **continuous prose**, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a)** Outline the reproductive cycle of HIV. [7]
- (b)** Explain the significance of the triple helix in the structure and function of collagen. [6]
- (c)** Define and explain the significance of the refractory period in nervous control. [7]

[Total: 20]

or

- 9 (a)** Explain how **one** named factor affects enzymatic activity in relation to protein structure. [7]
- (b)** Explain the significance of the population in natural selection. [6]
- (c)** Contrast the transmission of nerve impulse along a neurone and across synapse. [7]

[Total: 20]

END OF PAPER